EEES302 Digital Design

Experiment 3

Half Adder and Full Adder

1. Objective:
   - To design and analyze Half-Adder and Full-Adder circuits.
   - To physically implament these circuits on breadboard using basic logic gate IC.
   - To verify the theoretical truth table of these adder circuits by observing the physical output responses.

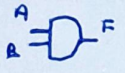
2. Review of experiment:

Half-Adder: It is adder circuit. It adds two single-bit binary inputs.

Full-Adder: It adds three single-bit binary numbers.

AND Gate

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1



OR Gate

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1



XOR Gate

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0



The adder circuits are digital logic circuits.

Half-Adder: The output terminals are called the sum and carry outputs. The sum output of a half-adder circuit is the exclusive OR (XOR) function of the two inputs.

Full-Adder: The full-adder circuit can be built by using two half adders with a third input called carry in.

Emrullah Yarar
2021513064

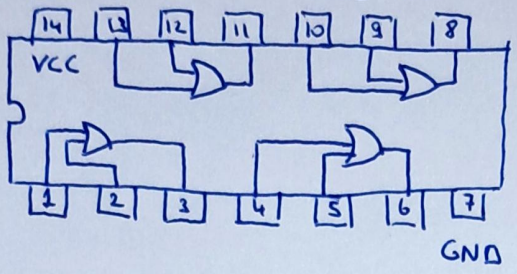
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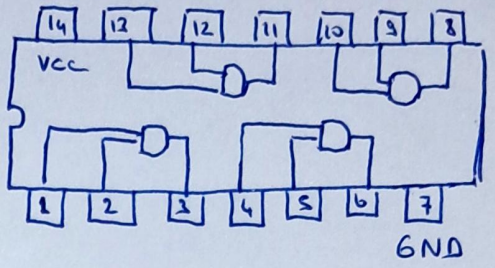
Half Adder and Full Adder.

3. Equipment List:

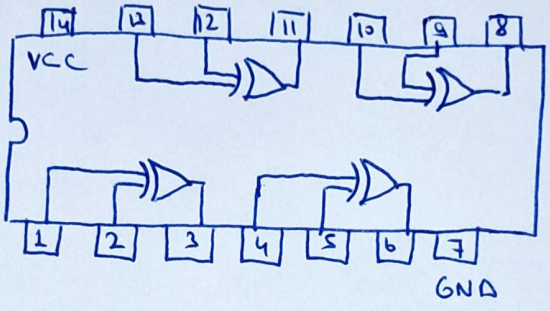
74LS32 TTL OR Gate IC



74LS08 TTL AND Gate IC



74LS86 TTL XOR Gate IC

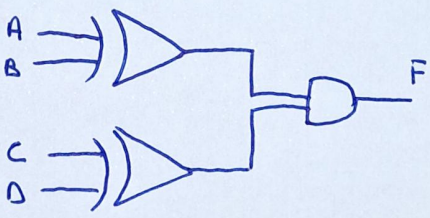


The IC's are includes 4 logic gates. The half and full adder circuits are built by using breadboard, jumper wires and logic gates.

4. Preliminary Work:

Implement the Boolean function $F = A\bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + \bar{A}B\bar{C}D$ with XOR and AND gates.

$F = (A \oplus B) \cdot (C \oplus D)$



The simplification done at preliminary sheet.

Emrullah Yeter
2021513064

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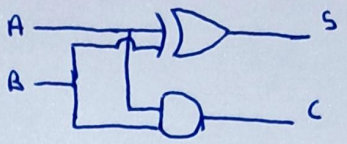
5. Experimental Procedure:

Half-Adder:

Truth Table:

A	B	Q	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic Gate Diagram:



Q = Sum C = Carry

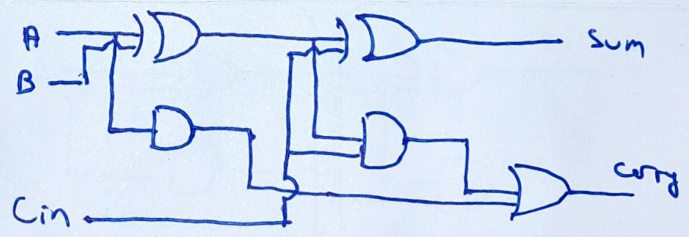
The Logic gate circuit is built at the breadboard by using 74LS86 and 74LS08. 1 gate is used for each IC. The input A and B are tested with logic high and low inputs. The output signal is measured from ossilator. The outputs controlled and gived the desired outputs from truth table.

Full-Adder:

Truth Table:

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Logic Gate Diagram:



The circuit is built by using same way.
74LS32 is used for AND gate IC.
Tested by using truth table and it gived the desired outputs.

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6. Experimental Result:

Half adder can build up by using one XOR gate and AND gate. It sums up two one binary inputs.

The result of sum is 1 for the either one of the inputs are logic high but not for both of them.

The result of carry is 1 for the both input logic high.

Full adder circuit can build up by using two XOR gate, two AND gate and one OR gate.

The circuit sums up three one bit binary numbers.

The experiment gived the output 1 for carry when the A and B input are logic high. In addition the sum become 0.

The experiment gived the output 1 for both Sum and carry for the input A=1, B=1 and C_{in} = 1.

The experiment done succesfully and the logic diagrams gived the correct outputs for the tested input values.

Emrullah Yaşar
2021513064